

CLOCK evolved in cnidaria to synchronize internal rhythms with diel environmental cues

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Abstract

The circadian clock enables anticipation of the day/night cycle in animals ranging from cnidarians to mammals. Circadian rhythms are generated through a transcription-translation feedback loop (TTFL or pacemaker) with CLOCK as a conserved positive factor in animals. However, the functional evolutionary origin and mechanism of action of CLOCK in basal animals are unknown. In the cnidarian *Nematostella vectensis*, pacemaker genes transcription including *NvClk* (the *Clock* ortholog) appears arrhythmic under constant light conditions, questioning the role of *NvCLK*. Utilizing CRISPR/Cas9, we generated a *NvClk* allele mutant (*NvClk*^{1/-}), revealing circadian behavior loss in constant light conditions (LL and DD) while a 24-hour rhythm was maintained under light-dark condition (LD). Transcriptomics showed distinct rhythmic genes in wild-type (WT) genes in LD compared to DD. The LD *NvClk*^{1/-} showed comparable numbers of rhythmic genes, whereas they were greatly reduced in DD. Furthermore, the LD *NvClk*^{1/-} showed alterations of temporal pacemaker genes expression, affecting their potential interactions. Additionally, differential expression of non-rhythmic genes associated with cell division and neuronal differentiation was observed. These findings suggest that while the light-responsive pathway can partially compensate for circadian clock disruption, the *Clock* gene has evolved in cnidarians to maintain 24-hour rhythmic physiology and behavior in constant conditions.

eLife assessment

This **fundamental** study for the first time defines genetically the role of the Clock gene in basal metazoa, using the cnidarian *Nematostella vectensis*. With **convincing** evidence, the study provides insight into the early evolution of circadian clocks. Clock in this species is **important** for daily rhythms under constant conditions, but not under a rhythmic light/dark cycle, suggesting that the major role of the circadian oscillator in this species could be a stabilizing function under non-rhythmic environmental conditions.

Introduction

Throughout the history of life on Earth, organisms have had to adapt to a constantly changing environment, including the ~24-hour daily rhythm of light/dark, driving the development of endogenous biological clocks. The circadian clock, which is entrained by external stimuli such as light, enables the organism to anticipate the onset of the light and dark phases and synchronize its physiology and behavior in harmony with the environment. This, in turn, enhances the organism's fitness and survival^{1,2,3}. From single-celled organisms to metazoans, circadian clocks have evolved multiple times, highlighting their importance to living organisms^{1,2,3}. Despite the fundamental role circadian clocks play in regulating the rhythmicity of living organisms, their evolutionary origin and intricate molecular mechanisms remain ambiguous in early diverging animal lineages, such as cnidaria.

Rhythmic phenomena, including calcification, reproduction, and diel behavior patterns, have been examined in cnidarian species^{3,4,5,6,7,8}. While some of these patterns were found to be directly triggered by environmental stimuli, such as light, others persist in the absence of external cues, suggesting the presence of an internally generated and self-sustaining circadian clock^{6,9,10,11}. At the molecular level, cnidarians possess homologs of putative core pacemaker genes found in bilaterians^{11,12,13}. Several studies have shown that most of these genes display diel expression patterns under light/dark cycles. However, in contrast to most animals, their oscillation generally ceases without the presence of light cues^{14,15,16}. Thus, how the core pacemaker genes orchestrate rhythmic gene expression and circadian behaviors in cnidarians remains unclear.

One of the most studied cnidarian species in the field of chronobiology is the estuarine sea anemone, *Nematostella vectensis*. Few studies have shown that in diel lighting, the locomotor behavior of *Nematostella* has a ~24-h rhythm that is maintained under constant conditions suggesting it is regulated by an endogenous circadian clock^{6,17,18,19}. In accordance, the *Nematostella* genome codes for conserved core pacemaker genes such as *NvClk*, *NvCycle*, and the cryptochromes *NvCry1a* and *NvCry1b*^{12,13}. The proposed circadian clock model in *Nematostella* is composed of the positive transcription factors (bHLH-PAS family), NvCLK and NvCYCLE, that heterodimerize and upregulate light-dependent cryptochrome genes in the feedback loop, and NvPAR-bZIPs in the feed-forward loop, which repress the transcription of the positive elements^{11,12,13,19}. More recently, the NvCLK-interacting pacemaker, NvCIPC was predicted to act as an additional repressor of the NvCLK:NvCYCLE dimer¹⁹. However, in contrast to the free-running oscillation demonstrated for *Nematostella* behavior^{6,17,19}, transcriptional expression profiles of most candidate genes implicated in the pacemaker do not retain their oscillation period in the absence of light^{14,15,19}.

Here, we employed the CRISPR/Cas9-mediated genome editing system to establish a *NvClk* mutant (*NvClk*^{1/Δ}) *Nematostella*. By combining behavioral monitoring and transcriptomic analysis, we aimed to elucidate the role of *NvClk* in regulating rhythmic locomotor activity and gene expression under varying light conditions. Our study revealed a robust light response pathway capable of compensation and a conserved function of CLOCK as a timekeeper in the absence of a light cue.

Results

Phylogenetic analysis and spatial expression pattern of *NvClk*

Phylogenetic analysis of NvCLK protein sequences revealed that *NvClk* is positioned within the cnidarian branch (Fig. 1a). It contains a basic helix–loop–helix (bHLH) DNA binding domain and two Per-Arnt-Sim (PAS) domains, similar to the protein structure found in other animals. PAS domains are crucial structural motifs involved in protein-protein interactions that drive the self-sustaining molecular mechanism underlying the circadian clock^{20,21}.

To localize *NvClk* expression at the polyp stage, *in situ* hybridization chain reaction (HCRv.3) was performed. Polyps were sampled at ZT10, i.e., peak expression of *NvClk*^{6,12,22}. *NvClk* specific expression was observed throughout the animal tissue, and enriched expression was visible in the tentacle endodermis and mesenteries, while no signal was observed in the negative control (Fig. 1b, Extended Data Fig. 1a). This expression pattern resembled the expression observed at the larvae stage²². To date, functional manipulation of the *NvClk* gene has not been performed in basal animal lineages including cnidarians, and its function is unknown in cnidarians^{20,21} (Fig. 1a).

Generation of *NvClk*¹^{-/-} *Nematostella*

To investigate the function of *NvClk* in *Nematostella*, we employed the CRISPR-Cas9 system to generate mutants. Based on existing knowledge from mouse and *Drosophila* models, we hypothesized that NvCLK:NvCYCLE dimer binds to the DNA motif CACGTG within the promoter of rhythmic target genes (Fig. 1c). Guide RNA (gRNA) was synthesized to target a region between the two PAS domains of the *NvClk* coding sequence (CDS). This gRNA, along with the Cas9 endonuclease, was microinjected into zygotes (Methods). Subsequently, F0 animals were outcrossed with wild type (WT), and the F1 progeny were raised to adulthood. Genotyping of F1 polyps identified 10 different mutated alleles, with six displaying a frame-shift mutation, including one with a 20 bp insertion (*NvClk*¹), resulting in a premature stop codon (Extended Data Fig. 1b). The predicted 203 amino acid truncated protein lacked 459 amino acids, including one co-factor dimerization PAS domain (Fig. 1c, Extended Data Fig. 1b). To obtain homozygous *NvClk*¹^{-/-} polyps, we crossed heterozygous *NvClk*¹ F1 animals. Genotyping of F2 polyps confirmed the expected 25% frequency of *NvClk*¹^{-/-} mutants. Subsequently, we intercrossed *NvClk*¹^{-/-} animals to obtain F3 *NvClk*¹^{-/-} polyps for use in subsequent experiments aimed at assessing the role of *NvClk* in regulating behavioral and genetic rhythms.

NvClk is necessary to maintain circadian behavior in constant conditions

To assess the impact of the *NvClk*¹ mutation on circadian rhythm, we monitored locomotor behavior of WT and *NvClk*¹^{-/-} polyps under different light conditions (Supp. Table 1). Both the WT and *NvClk*¹^{-/-} populations exhibited a 24-hour periodicity in 12:12 hour Light-Dark (LD) cycles (Fig. 1d,e), with 15 out of 21 WT animals displaying 24-hour rhythmicity compared to only 9 out of 20 *NvClk*¹^{-/-} animals (Fig. 1f, Table 1, Supp. Table 2). The average acrophase for WT polyps (13.3 hours) was significantly lower than for *NvClk*¹^{-/-} polyps (17.3 hours) (Fig. 1f, Table 1).

We then investigated locomotor behavior under continuous conditions, namely continuous dark (DD) or continuous light (LL). WT polyps exhibited a 22-hour rhythmic behavior under both constant light conditions, with 17 out of 25 WT polyps displaying a 24-hour rhythm under DD and 7 out of 25 under LL (Fig. 1g-i). In contrast, few *NvClk*¹^{-/-} polyps displayed rhythmic behavior

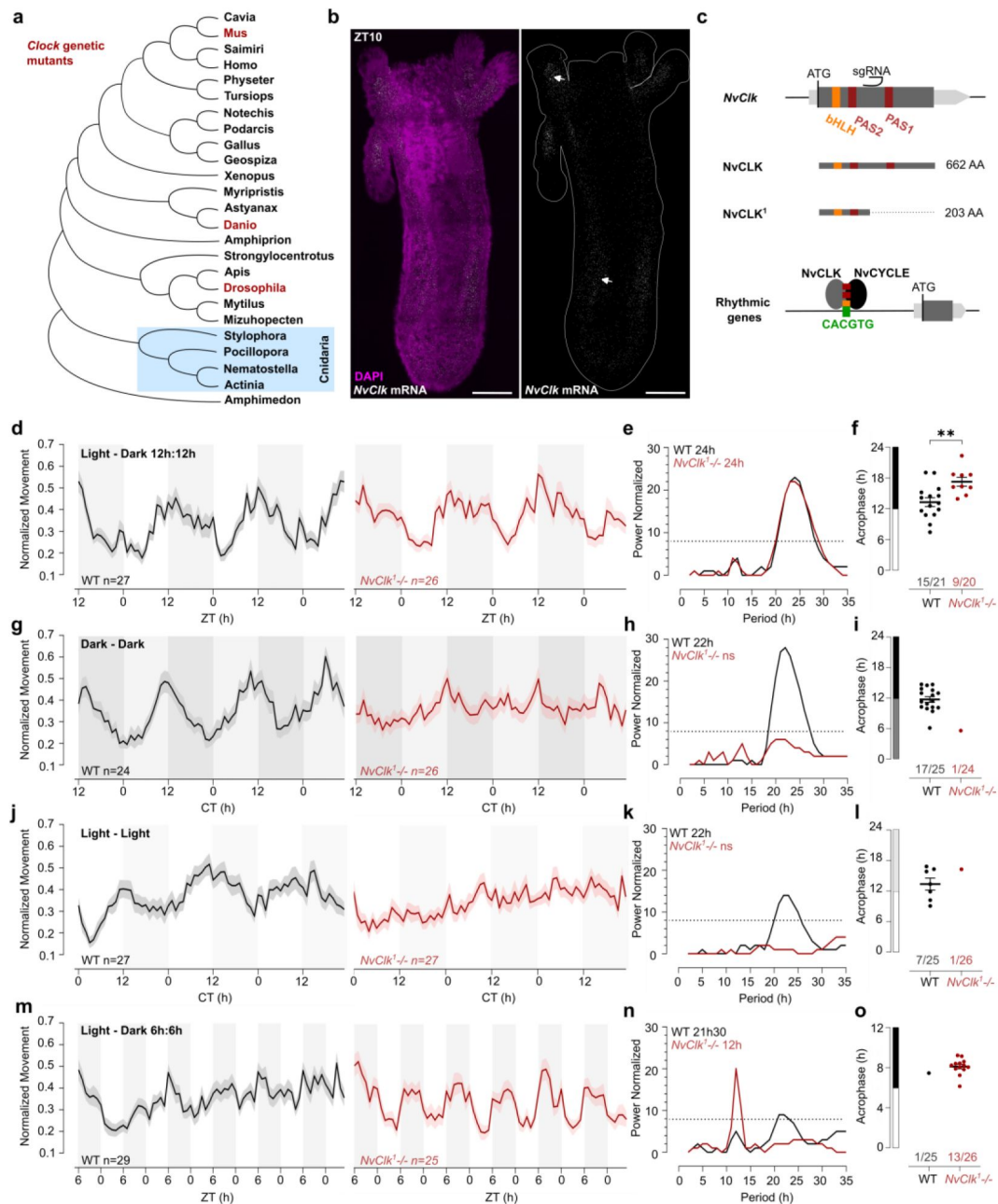


Fig. 1

NvClk^{1-/-} cannot maintain circadian behavior in non-diel light conditions.

(a) Phylogenetic tree showing the evolutionary relationship of CLK orthologs across different animal species. (b) *In situ* hybridization of *NvClk* in the WT juvenile with scale bars representing 0.1mm. (c) Schematic representation of the *NvClk* gene in grey, with the open reading frame (ORF) in dark grey and the conserved protein domains bHLH (yellow) and PAS1 and PAS2 (dark red). The CRISPR generated *NvClk*^{1-/-} allele has a +20nt insertion after the PAS1 domain, represented by a black arrowhead. NvCLK dimerizes via its three functional domains with NvCYCLE binding the CACGTG ebox to drive rhythmic transcription. (d-g-j-m). Normalized Movement (a.u.), hourly binned over 72h, under different light conditions: 12h light:12h dark, continuous dark (Dark - Dark), continuous light (Light - Light), and 6h light:6h dark. The black line represents the WT and the red line represents the *NvClk*^{1-/-} mutant. (e-h-k-n) Lomb-Scargle Periodograms for each corresponding light condition. The significant period value (p<0.01) is indicated for each genotype in the top left corner of each graph. (f-i-l-o) Phase detection (Cosinor) and genotype comparison of 24h rhythmic individuals. See number nrhythmic/ntotal on the x-axis indicating for each genotype the number of 24h-rhythmic animals over the total population.

24-hour period	Cosinor		
p.adj<0.01	WT	<i>NvClk1+/-</i>	<i>NvClk1-/-</i>
LD	15/21	-	9/20
DD	17/24	9/19	1/25
LL	7/25	-	1/26
LD66	2/28	-	0/25
DD66	1/27	-	0/24
12-hour period	Cosinor		
p.adj<0.01	WT	<i>NvClk1+/-</i>	<i>NvClk1-/-</i>
LD	1/21	-	0/20
DD	0/24	0/19	0/25
LL	0/25	-	0/26
LD66	1/28	-	13/25
DD66	0/27	-	0/24

Table 1.

Summary of rhythmic analysis of individual behavior.

under constant conditions (1 out of 24 in DD and 1 out of 26 in LL) (**Table 1**). Additionally, we observed an intermediate phenotype in the locomotor behavior of heterozygous polyps for the *NvClk¹* allele in DD (**Extended Data Fig. 1c-f**).

To further explore the 24-hour rhythm of *NvClk¹* polyps under LD, we tracked locomotor activity under a 6-hour light: 6-hour dark (LD 6:6) cycle after a regular diel 72-hour entrainment under 12:12 LD. While WT polyps maintained a marginally significant periodicity of 22 hours, *NvClk¹* polyps displayed a 12-hour rhythm at the population level (**Fig. 1m-o**). Specifically, we identified an important difference of 12-hour rhythmic individual polyps between WT and *NvClk¹* groups (1 out of 25 WT polyps vs. 13 out of 26 *NvClk¹* polyps) (**Table 1**). Notably, entrainment with LD 6:6 did not lead to a 12-hour rhythm in DD for both WT and *NvClk¹* polyps (Extended Data Fig. 1g-i).

***NvClk* regulates rhythmic gene expression differentially in response to light conditions**

To investigate the underlying molecular correlates of the behavioral phenotype found in *NvClk¹* polyps, we conducted transcriptional profiling. WT and *NvClk¹* polyps were sampled seven times every 4 hours over 24 hours under LD and DD conditions (**Fig. 2a**). We defined true rhythmic genes using a combination of both RAIN (rhythmicity analysis incorporating non-parametric methods) and JTK_CYCLE algorithms (**Fig. 2b**, Supp. Table 3). Within our dataset, we identified 119 genes rhythmic under LD and 107 rhythmic genes under DD in WT polyps (**Fig. 2b**). In *NvClk¹* polyps, we detected 147 rhythmic genes under LD and only 37 under DD (**Fig. 2b**).

The rhythmic genes in WT polyps displayed a higher acrophase under DD compared to LD (17.20h vs 12.93h, **Fig. 2c**). However, no differences were detected between LD and DD rhythmic genes in *NvClk¹* polyps (**Fig. 2d**). Similarly, the relative amplitude (rAMP) of DD rhythmic genes was higher in WT polyps compared to LD (0.61 vs 0.43, **Fig. 2e**), but no rAMP difference was observed between LD and DD rhythmic genes in *NvClk¹* polyps (**Fig. 2f**).

We performed a clustering analysis on the rhythmic genes using the DPGP model (Dirichlet process Gaussian process mixture model). The average number of genes per cluster between LD and DD conditions in WT polyps did not differ significantly (7.3 vs 7.6, **Fig. 3a**, Supp. Table 4). However, in *NvClk¹* polyps, the average number of genes per cluster was significantly lower in DD compared to LD (4.1 vs 8.6, **Fig. 3b**). Specifically, in WT DD polyps, we observed clusters with high numbers of genes peaking at subjective night (**Fig. 3c**, Supp. Table 4). We did not identify any GO term enrichment to any of the clusters.

***NvClk* regulates temporal expression pattern of pacemaker genes**

In line with previous findings in *Nematostella*^{12,14}, candidate pacemaker genes showed arrhythmic expression under DD conditions (**Fig. 4a**, Supp. Table 3). However, the altered expression patterns observed in LD *NvClk¹* polyps, compared to LD WT polyps (**Fig. 4a**, Supp. Table 3), suggest a complex regulatory effect.

To systematically analyze the mutation's impact on these genes, we constructed a correlation matrix based on their expression levels. In LD WT polyps, genes clustered into two groups: one with *NvClk* peaking during the day, anticorrelated to the second cluster (*NvPar-bzipc* and *NvCipc*) peaking at night. In LD *NvClk¹* polyps, the second cluster contained two additional genes and showed weaker anticorrelation with the *NvClk* cluster (**Fig. 4b**).

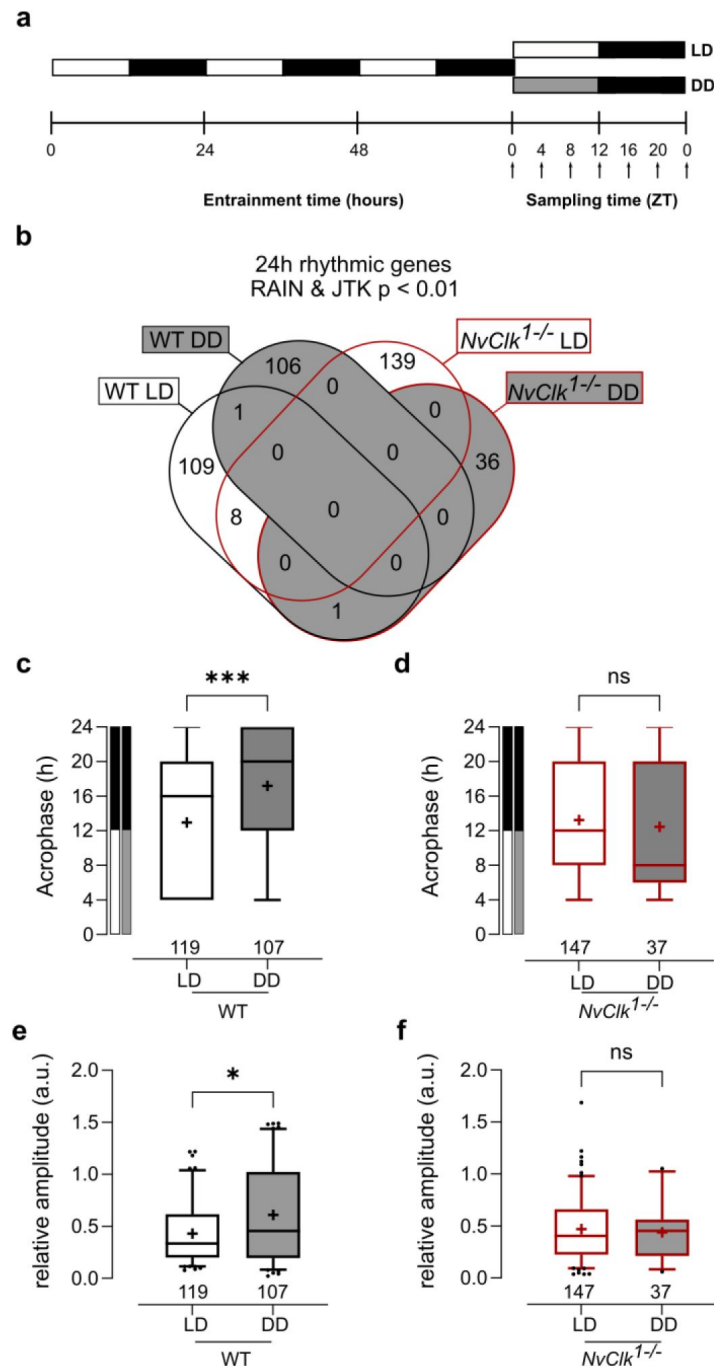


Fig. 2

NvClk^{1-/-} shows rhythmic genes reduction in constant darkness with altered rhythmic features.

(a) Overview of the experimental design used to generate RNA-seq data. Polyps were entrained for 72 hours before sampling at 4-hour intervals over a 24-hour period (dark arrows) in both LD and DD cycles. (b) Venn diagram comparing the total number of 24h rhythmic genes identified in WT and *NvClk*^{1-/-} in LD and DD cycles with a $p < 0.01$ with RAIN and JTK. (c) Average acrophase comparison between rhythmic genes in LD and DD in WT polyps. Mann-Whitney test, $p < 0.001$. (d) Average acrophase comparison between rhythmic genes in LD and DD in *NvClk*^{1-/-} polyps. Mann-Whitney test, $p > 0.05$. (e) Average relative amplitude comparison between rhythmic genes in LD and DD in WT polyps. Mann-Whitney test, $p < 0.05$. (f) Average relative amplitude comparison between rhythmic genes in LD and DD in *NvClk*^{1-/-} polyps. Mann-Whitney test, $p > 0.05$. (c-f) sample size (n) indicated below each boxplots.

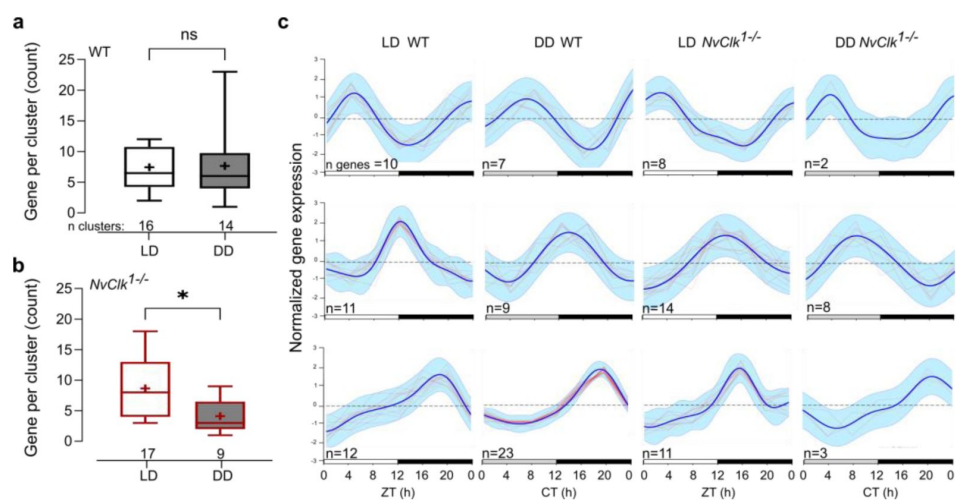


Fig. 3

***NvClk1^{-/-}* shows temporal gene clusters recruitment alteration in constant darkness.**

(a) Gene per cluster count comparison between LD and DD in WT polyyps. Number of cluster indicated below each boxplot. Mann-Whitney, $p > 0.05$. (b) Gene per cluster comparison between LD and DD in *NvClk1^{-/-}*. Sample size (n) indicated below each boxplot. Mann-Whitney, $p < 0.05$. (c) Representative temporal gene cluster for each condition. For both light conditions, from top to bottom cluster peaking at day, then twilight, then at night. Number of gene for each specific cluster indicated on the x-axis.

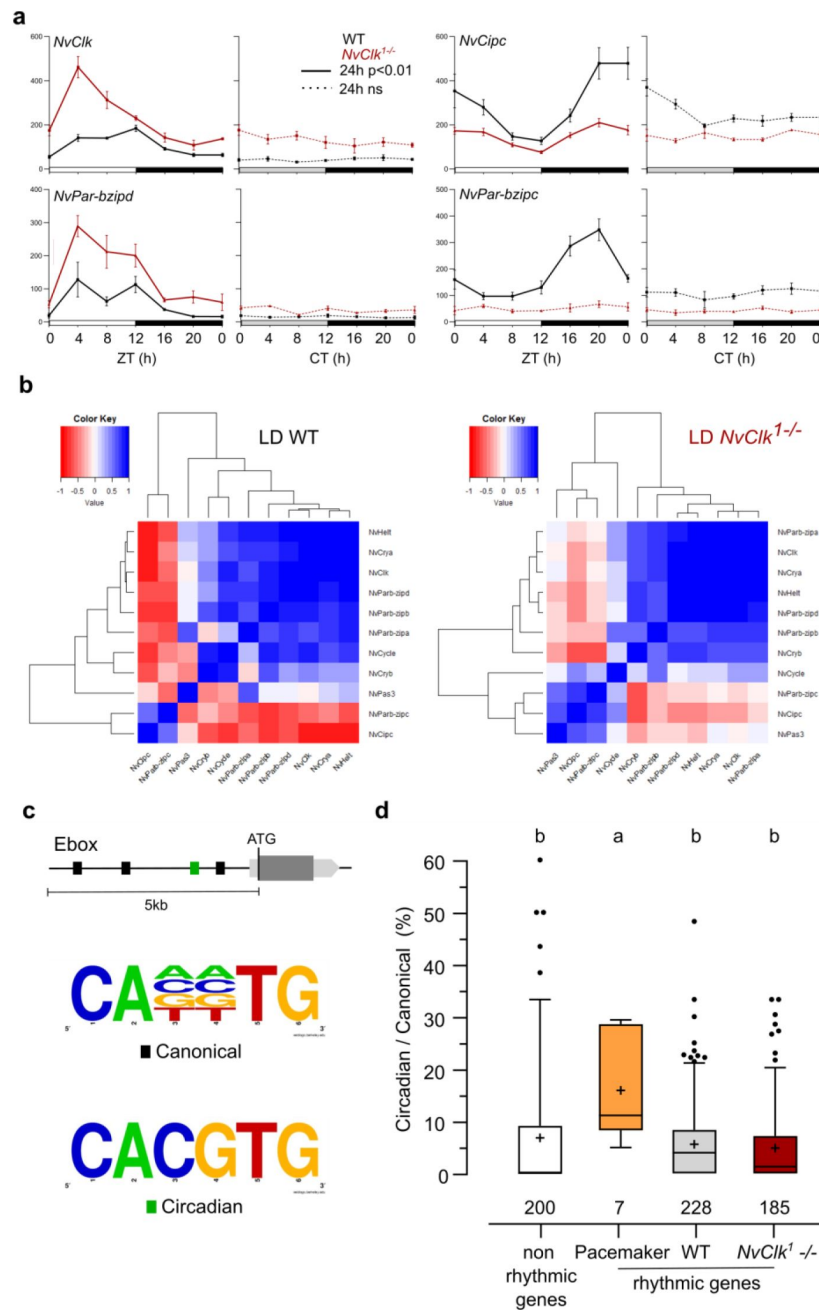


Fig. 4

NvClk^{1-/-} alters temporal pacemaker genes expression.

(a) Four pacemaker gene are individually plotted showing the read count over 24h in LD and DD in WT (black) and *NvClk*^{1-/-} (red). The continuous line represents significant rhythmicity (RAIN&JTK p<0.01) while the dashed line indicates no rhythmicity. (b) correlation matrix of candidate pacemaker genes expression in LD for WT on the left and *NvClk*^{1-/-} on the right. (c) schematic representation of the promoter sequences analyses 5kb upstream the putative ATG. Black boxes represent canonical E-boxes while circadian E-boxes are green. Below the logo motif we used to identify canonical and circadian Ebox. (d) Circadian / Canonical ratio (in %) per condition. Kruskal-Wallis, multiple comparison, a vs b : p<0.05.

To go further into the regulatory mechanisms downstream of the pacemaker, we examined the presence of circadian E-box motifs (CACGTG) within 5kb upstream of the predicted ATG of rhythmic genes. We calculated circadian/canonical E-box enrichment to account for the total variation in the number of canonical E-boxes (**Fig. 4c**). Notably, only the candidate pacemaker genes exhibited a significant enrichment in circadian E-boxes in their promoters (15.9%) compared to the WT (5.6%), *NvClk*^{1^ΔΔ} (4.8%) rhythmic genes, and non-rhythmic genes (6.8%) (**Fig. 4d**).

***NvClk* coordinates cell division and neuronal pathways in constant darkness**

In addition to the transcriptomic rhythmic analysis, our aim was to identify processes regulated by *NvClk* that may not necessarily exhibit rhythmicity. To achieve this, we conducted a differential gene expression analysis on the total transcriptome between genotypes under each light condition. Under LD conditions, *NvClk*^{1^ΔΔ} polyps exhibited 457 down-regulated genes and 646 up-regulated genes, with no significant enrichment in GO terms observed (**Fig. 5a**, Supp. Table 4 and 5). However, in DD conditions, *NvClk*^{1^ΔΔ} displayed 2450 down-regulated genes and 1770 up-regulated genes (**Fig. 5b**, Supp. Table 4). Notably, the down-regulated genes were predominantly enriched in processes related to mitosis, microtubules, and ciliary/flagellar motility. Conversely, the up-regulated genes showed significant enrichment in processes such as the modulation of another organism's processes, axonal guidance, and sensory perception (**Fig. 5b**, Supp. Table 5).

Discussion

Conserved behavioral CLOCK function through animal evolution

Our study provides valuable insights into the evolution of circadian clocks by characterizing the effects of the first *Clock* mutation in a cnidarian, the sea-anemone *Nematostella vectensis*. Through our behavioral assays, we have shown that *NvClk* is essential for maintaining rhythmic locomotor activity in the absence of an entraining light cue. Although the rhythmicity of the *NvClk*^{1^ΔΔ} heterozygote polyps was affected in DD, our results could not discriminate a dominant-negative from a full loss of function to identify the nature of this mutation (Extended Data Fig. 1g-i). Studies in various model organisms further support the importance of CLOCK in regulating circadian locomotion. For instance, homozygous *DmClk*^{-/-} flies exhibit a loss of circadian locomotion in constant darkness^{23,24}, while the *DnClk1a* zebrafish mutant displayed a shortened period under the same conditions²⁵. Interestingly, *MmClock*^{-/-} mice maintain robust circadian locomotion in constant darkness, suggesting potential functional redundancy among mammalian clock genes²⁶⁻²⁹. Overall, these findings support a conserved role of CLOCK in preserving circadian behavioral rhythms in absence of light cues across the distant *Nematostella*, flies, zebrafish, and mice.

Moreover, the conservation of a 24-hour locomotion rhythm in LD of the *NvClk*^{1^ΔΔ} polyps with a different mean acrophase, revealed a light-response pathway independent of the circadian circuit, consistent with observations in other animal models^{23,26,30} (**Fig. 1f**). *NvClk*^{1^ΔΔ} polyps exposed to a 12:12h LD cycle exhibited a 24-hour period, whereas those exposed to a 6:6h LD cycle displayed a 12-hour period. Notably, nearly no WT polyps exhibited a 12-hour rhythm under this condition, suggesting that the circadian clock overrides the light-response pathway (**Fig. 6a**). While some of the circadian factors can directly sense the light such as CRY proteins³¹, 29 putative *NvOpsin* have been identified in the genome which could be involved in the light-response pathway³². Behavioral tracking of *NvClk*^{1^ΔΔ} polyps exposed to different wavelength could help to identify candidates for further functional studies of the light-response pathway.

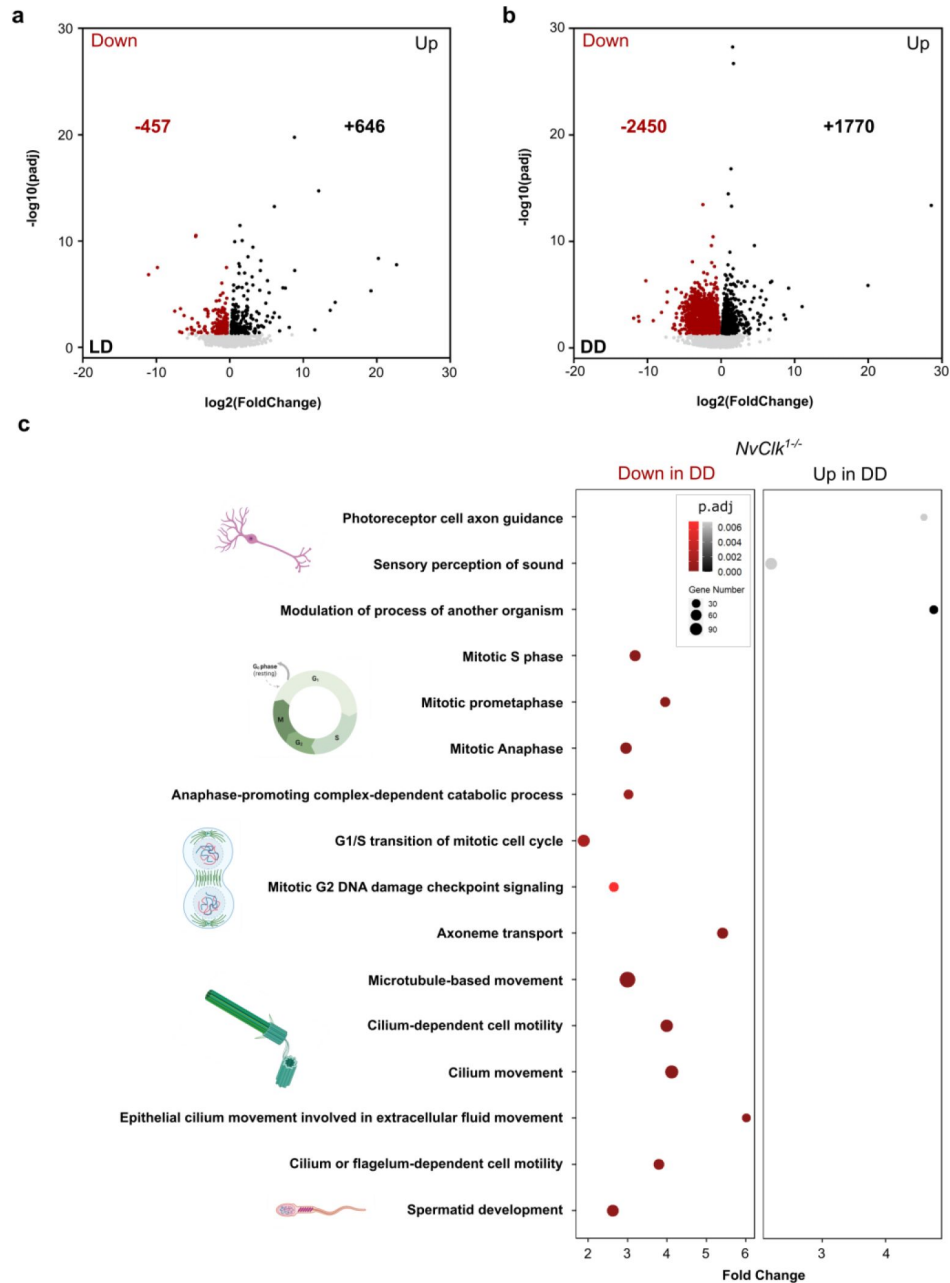


Fig. 5

NvClk1^{-/-} disrupts cell-cycle and neuronal pathways in constant darkness.

(a) Volcano plot showing the differential expression of genes (DEG) between WT and *NvClk1*^{-/-} in LD (Left) and DD (right). Dashed line indicate the threshold used to detect DEG ($\text{p.adj} < 0.01$). Red dots indicate down regulated genes and black dots up-regulated genes in *NvClk1*^{-/-} compare to WT polyps (b) Gene Ontology (GO) terms with with significant fold-enrichment (Bonferroni corrected p-value or $\text{p.adjusted} < 0.01$) for the DEG analysis in DD. Down regulated genes in Red while Up regulated genes in Black.

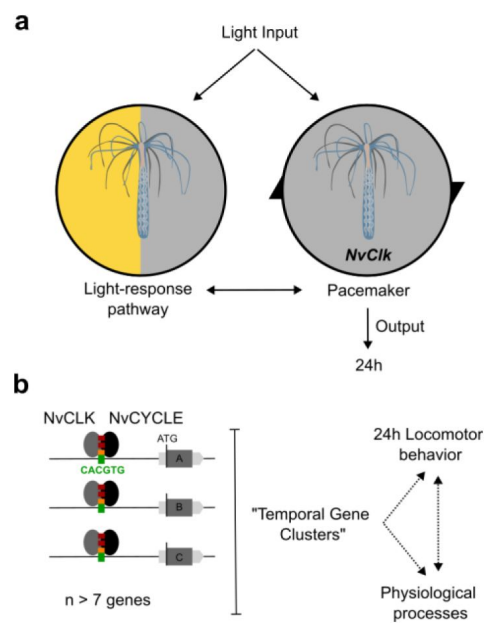


Fig. 6

Summary of *NvClk* function in the regulation of *Nematostella* circadian rhythmicity.

Transcriptional rhythmicity plasticity downstream *NvClk*

At the transcriptomic level, previous studies in *Nematostella* have shown large changes in the transcriptional profile of many genes after a single day of constant darkness, including the candidate pacemaker genes that were found arrhythmic, despite sustain circadian locomotion^{12,14,22}. Consistent with previous transcriptomic analysis in cnidarian^{7,14,16,33}, most of rhythmic genes identified in LD were different from those identified in DD in the WT polyps. Notably, they displayed higher mean acrophase and larger mean amplitude in DD suggesting a differential regulation in response to light conditions, which has not been investigated in previous cnidarian studies. Additionally, an overlap between our LD rhythmic genes and Leach et al. dataset¹⁴ suggests the robustness of pacemaker rhythmic transcription in LD conditions compared to downstream rhythmic genes (Extended data Fig. 2). Overall, these observations suggest a potential plasticity of the pacemaker in selecting specific genes depending on environmental conditions.

Similar to the rhythmic behavior of *NvClk*^{1-/-} polyps under LD conditions, the identification of 24-hour rhythmic genes suggests potential compensation by the light-response pathway. However, most of rhythmic genes in the *NvClk*^{1-/-} compared to WT polyps in LD conditions are different. This suggests the light-response pathway may not fully reproduce the normal circadian pattern observed in WT polyps and would require further investigation to understand their recruitment and function. Furthermore, the identification of a reduced number of rhythmic genes in DD *NvClk*^{1-/-} polyps revealed the importance of *NvClk* in maintaining molecular rhythm in absence of light. However, the persistence of these rhythmic genes raises questions regarding their origin. They could be the result of the mutation itself, a compensatory mechanism or false positives resulting from our analysis. The study by Ray et al., which report controversial findings regarding RNA rhythmicity in the *Bmal1*^{-/-} mouse cells (*NvCycle* orthologous in *Nematostella*) is an interesting case of rhythmicity analysis to consider³⁴⁻³⁸. Our attempt to identify rhythmic genes faced challenges, particularly in the choice of statistical parameters. Initially, using stringent criteria like BHQ for the JTK method or p.adj with RAIN resulted in a very limited number of genes for analysis (Supp. Table 3). Consequently, we opted for a balanced approach combining both algorithms with a p<0.01 threshold. We made a compromise to identify rhythmic genes with greater confidence, but we are aware that the methodological choices are critical for the results.

The clustering analysis revealed that rhythmic genes can be categorized by expression pattern similarity. This approach defined “temporal gene clusters” as group of seven/eight genes in average in the WT (Fig. 6b). Their discovery poses a fundamental question: How a group of genes are co-regulated in times and space (cell types) by the pacemaker? Their recruitment is disrupted in the DD *NvClk*^{1-/-} polyps suggesting an essential function of *NvClk* in absence of light. The combination of published scAtlas³⁹ and multiplexed FISH techniques⁴⁰ will be essential to further investigate the biological regulation and function of these temporal gene clusters.

NvClk temporally organize pacemaker genes expression to drive rhythmic genes recruitment

Our study reveals that *NvClk* plays a key role in regulating the temporal transcription of pacemaker candidate genes (Fig. 4a). Our analysis identified two clusters of pacemaker genes: One containing *NvClk*, and a second one containing a potential *NvClk* inhibitor (*NvCipc*)^{41,42}. The alteration of clusters composition with a weaker anticorrelation in LD *NvClk*^{1-/-} polyps might generate a desynchronization of the pacemaker factors availability. Indeed, regulation of rhythmic transcription involved a complex protein-protein-DNA timing interaction. Furthermore, we did not identify any circadian E-boxes enrichment in rhythmic genes between conditions, except for

the candidate pacemaker genes. Altogether, this supports the function of *NvClk* in orchestrating the timing interaction of pacemaker factors to select downstream rhythmic genes, indicating a more complex regulatory landscape at play.

However, one significant unanswered question in our study is the reason for the arrhythmic transcription of putative pacemaker genes in DD. We hypothesize that using whole animals for sampling material might mask oscillating gene expression signals, especially if signals are present in a small number of cells or if tissues exhibit rhythmic gene expression in different phases. Furthermore, it is important to acknowledge a limitation in our interpretation, common in chronobiology: the use of RNA oscillation as a proxy for protein oscillation and function. The development of tools to study the pacemaker factors at the protein level in *Nematostella* will leverage this limitation in the field.

***NvClk* regulates processes involved in cell proliferation and the neural system in absence of light**

Our study of *NvClk* suggests coordination of cellular processes, especially in the absence of light. The results of our rhythmic transcriptomic analysis (Fig. 2 [↗](#) [↗](#)) raised questions regarding indirect effects, and the non-rhythmic function of *NvClk*. We performed a differential gene expression analysis on the total transcriptome for each light condition. Under LD conditions, *NvClk*¹[↗](#)^{-/-} polyps exhibited significant changes in gene expression but no GO term enrichment was found (Fig. 5a [↗](#), Supp. Table 5) revealing multiple altered processes we cannot yet identified. In contrast, under DD conditions, *NvClk*¹[↗](#)^{-/-} polyps displayed more pronounced alterations, with more DEGs and enriched GO-terms for down-regulated genes related to mitosis, microtubule organization, and ciliary/flagellar motility, while the up-regulated genes showed enrichment in processes such as the modulation of other organism's processes, axonal guidance, and sensory perception (Fig. 5b [↗](#), Supp. Table 5). These results imply that *NvClk* has non-circadian functions that are dependent on light availability. This is particularly noteworthy considering the expression of core pacemaker genes, which are known to be arrhythmic during larvae stages, potentially involved in developmental processes²²[↗](#).

Taken as a whole, this study provides novel insights into circadian regulation in *Nematostella vectensis* and sheds light on the evolutionary origin of circadian time maintenance. Our findings indicate that CLOCK function is conserved from Cnidaria to mammals to maintain rhythmicity in absence of light cues. Furthermore, it revealed a light-response pathway able to compensate at both behavioral and molecular level using light cues. This circadian clock mutant opens new avenues for investigating cell-type-specific mechanisms of the circadian clock that drive the molecular and phenotypical oscillations of cnidarians. By further exploring the circadian clock mechanisms in cnidarians, we can gain deeper insights into the evolutionary origins of this critical aspect of biology, enhancing our understanding of how organisms have evolved to keep track of time and adapt to their environment.

Methods

Animal husbandry

Nematostella were grown in 12g.L sea salt water at 17°C, and fed with *Artemia salina nauplii* three times a week. Spawning of gametes and fertilization was performed according to a published protocol⁴³[↗](#). In brief, temperature was raised to 25 °C for 9 h and the animals were exposed to strong white light. Three hours after the induction, oocytes were mixed with sperm to allow fertilization.

CRISPR/Cas9 mediated mutagenesis

Genome editing in *Nematostella* was carried out following established CRISPR/Cas9 protocols, with slight modifications^{44,45}. ZiFiT targeting software (<http://zifit.partners.org/>)⁴⁶ was used to select a guide RNA (gRNA) target site within the beginning of the *NvClk* exon and to design complementary oligonucleotides. To ensure specificity of the gRNA, the selected target site sequence (GGTCCTCTCGTGGACTCTAC) was BLASTed against *Nematostella vectensis* genome (using JGI expected E value threshold of 0.1 to adjust for short sequences: <http://genome.jgi.doe.gov/Nemve1/Nemve1.home.html>)⁴⁷. To generate the gRNA template, the following oligonucleotides were used:

Oligo 1: 5'- TAGGTCCTCTCGTGGACTCTAC -3'

Oligo 2: 5'- AAACGTAGAGTCCACGAGAGGA -3'

To construct the gRNA expression vector, pDR274 (plasmid # 42250; Addgene) was digested with the *BsaI* restriction enzyme. Subsequently, the gRNA oligonucleotides were annealed and cloned into the *BsaI*-digested pDR274 vector. Next, *DraI*-digested gRNA expression vectors, purified via ethanol precipitation followed by PureLink PCR purification kit (Invitrogen), were transcribed and purified using HiScribe T7 High Yield Transcription Kit (New England BioLabs) and illustrated using Microspin G-50 Columns (GE Healthcare Life Sciences), respectively. Cas9 recombinant protein with nuclear localization signal (260 ng/μl; PNA Bio, USA) was co-injected with the gRNA (140ng/μl) into *Nematostella* zygotes. Injected embryos were raised in petri dishes at 22°C under constant darkness with daily water changes.

CRISPR/Cas9 mediated mutagenesis screening

To evaluate genome editing efficiency and mosaicism in F0 animals, genomic DNA flanking the guide sequence was amplified and Sanger sequenced. PCR was performed using two strategies. For the first, PCR reactions were performed using individual injected *Nematostella* (7-days post-fertilization), directly pipetted into a 25- μl PCR reaction containing a 2x concentration of PCR MasterMix (Tiangen) and 10 pmol of each PCR primer. For the second, genomic DNA was extracted from tissue sampling from live adult animals, after relaxation in 7% MgCl₂ (Sigma-Aldrich), using NucleoSpin Tissue DNA purification kit (MACHERY-NAGEL). Subsequent PCR reactions were performed as above using 50-ng of genomic DNA. The primers used for these reactions (listed below) were designed to amplify a ~750bp region around the targeted *NvClk* genomic locus. Mosaicism was determined if sequenced PCR products showed overlapping peaks in their chromatograms. The second strategy, which takes advantage of the ability of *Nematostella* to fully regenerate within a few days^{47,48}, is the one we refer to in the text hereafter. The injected individuals determined mosaic mutants were raised as F0 founders to sexual maturity and outcrossed with wild-type animals. The progeny of these crosses was raised and individually genotyped as described above. To determine inheritable mutations, sequences were further analyzed using the Tracking of Indels by DEcomposition web-tool (TIDE). TIDE quantifies editing efficiency and identifies the predominant types of DNA insertions and deletions (indel) mutation composition from a heterogeneous PCR product compared to a wild-type sequence⁴⁹. Different heterozygous mutants were raised to sexual maturity and outcrossed with wild-type animals. The resulting F2 progenies were then raised to sexual maturity and genotyped before spawning for F3. Heterozygous mutants from each F2 progeny were intercrossed to obtain 25% homozygous F3 mutants. All animals used in this study are derived from heterozygous F2 mutants intercrosses, harboring the mutant allele *NvClk*¹. PCR genotyping was performed using the following primer:

Forward 5'- GATAAACACGGGCCGAAGATA -3'

Reverse 5'- CAGTCCACGCTGGTCTAAAT -3'

Determination of *NvClk*¹ F3 mutant genotypes

Genomic DNA was extracted as described above and used for following PCR and electrophoresis-based genotyping. PCR primers (listed below) encompassing the *NvClk* targeted site were used to produce PCR products of approximately 100 bp. The PCR products were then loaded and migrated by electrophoresis on a 3% Tris-borate- EDTA (TBE) agarose gel supplemented with GelStar Nucleic Acid Gel Stain (Lonza) for approximately one hour. The genotype of each F3 animal was determined by visualizing differences in migration speed of the PCR products caused by the CRISPR/Cas9 genome editing. The homozygous mutant animal (*NvClk*¹^{-/-}) produces only the larger ~120 bp amplicon while the wild-type animal (*NvClk*¹^{+/+}) produces only the lower ~100 bp amplicon. Animals heterozygous for the deletion (*NvClk*¹^{+/-}) produce both the larger mutant and the smaller wild-type amplicons. PCR genotyping was confirmed by subsequent DNA sequencing of selected F3 animals.

PCR was performed using the following primer:

Forward 5'- ACCCACTGAGTGACCTCTT -3'

Reverse 5'- ATACGCCTGCGCTATACACC -3'

Behavioral assays

Locomotor activity of individual *Nematostella* were monitored using a lab-made setup equipped with an IP Infra-Red camera (Dahua Technology, Hangzhou, China), a white neon illumination (Aquastar t8, Sylvania Lightning Solution,) and constantly illuminated with low-intensity infrared (850nm) LED light. The camera output 1h mp4 movie files which were AVI converted then stitched. The data collection and analysis were carried out by EthoVision XT8 video tracking software (Noldus information technology, Wageningen, Netherlands). Animals were isolated in wells of six-well plates, each of which was manually defined as a tracking 'arena' in the EthoVision software. Center- point detection with gray scaling (detection range of 25–77, contour erosion of 1-pixel, high pixel smoothing) was used to monitor movements, which were calculated according to the change in position of the average center pixel each second.

Illumination was provided with an intensity of 17 PPFD (+/-2) and did not significantly affect the experimental temperature (20 °C). The illumination cycles were 12: 12 h Light-Dark, 6: 6h Light-Dark or continuous light (LL). Parameters were optimized to ensure that organisms were detected throughout the entire observation period.

Behavior analysis

The total distance moved was summed in hourly bins and individually normalized min/max by the software GraphPad Prism 9.4. The average and standard errors were calculated for all tested animals based on the normalized values of each hour. The oscillation frequencies of the average population were evaluated based on the average values of each experiment using Fourier analysis-based software LSP with a $p < 0.01$. For individual analysis we used the online platform Discorhythm, combining different algorithms including Cosinor, JTK and LSP⁵⁰. We chose Cosinor it is the one designed to detect efficiently the acrophase.

RNA-seq experimental design

All polyps were isolated in wells of six-well plates. Then, they were subjected to the 12: 12h LD cycle with 17 PPFD (+/-2) light intensity during 72 hours for entrainment in an incubator with a stable temperature at 18°C. Subsequently, the polyps were divided into 2 experimental subgroups: 12: 12h LD and Continuous Dark or DD. Sampling began at 7 am (ZT0) and was performed at 4-

hour intervals over 24 hours. At each time point, three or four individual polyps were sampled from each experimental group, immediately snap-frozen in liquid nitrogen, and transferred to -80°C for storage.

RNA extraction, library preparation, and sequencing

Total RNA was extracted from all sampled polyps ($n = 96$) using TRIzol reagent (Invitrogen). Purified RNA samples were analyzed using a NanoDrop 1000 spectrophotometer (Thermo Fisher Scientific) to assess RNA quantity and 2200 TapeStation (Agilent) to assess RNA quality (RNA integrity number, >8.5). From each of the 96 samples (4 biological replicates in LD and 3 in DD, with the highest-quality extracts $\times 4$ experimental subgroups $\times 7$ time points, $1.5\text{ }\mu\text{g}$ of RNA was sent for library preparation (INCPM mRNAseq protocol) and sequencing at the Weizmann Institute Sequencing Unit, Israel. The libraries were sequenced using the bulk MARS-seq protocol (Jaitin et al. Science, 2014, Keren-Shaul et al., Nature Protocols, 2019) on an Illumina NovaSeq 6000, resulting in an average of 17 million single-end reads of 113 bases per sample.

Bioinformatic analysis

First, the unique molecular identifier (UMI) sequence of each read was extracted and placed within the read1 header file using UMI-tools extract (umi_tools v1.1). Next, the reads were mapped onto the *Nematostella* genome (NCBI genome GCA_000209225.1) using STAR (v2.6.0a) (Dobin et al., 2013) with default parameters. Mapped reads were then deduplicated based on UMIs using the umi_tools dedup. The mapped reads were sorted by SAMtools (version 1.9). The number of reads per gene were quantified using HTSeq-Count (v0.12.4) (Anders et al., 2015).

Rhythmicity analysis

Rhythmicity in transcript expression was assessed using the RAIN (ref-23) and metacycle (Wu et al., 2016) packages in R. The RAIN and JTK algorithms from metacycle were run separately for each *Nematostella* genotype in both light conditions (LD and DD), treating them as individual datasets. All replicates ($n=3$) for each time point within a dataset were analyzed as regular time series to identify transcripts exhibiting daily oscillations. Specifically, we focused on transcripts with a precise 24-hour period, excluding those with a range (e.g., 10 to 14 or 20 to 28 hours). To improve the accuracy of identifying true rhythmic genes, only transcripts with a P value < 0.01 in both RAIN and JTK analyses were deemed confidently cycling transcripts. Genes identified as significant cycling genes were subsequently utilized as input for the DPGP_cluster program (McDowell et al., 2018), which clusters genes based on their expression trajectories. Gene clusters comprising 10 or more genes underwent testing for GO term enrichment. Heatmaps were generated using the heatmap package (v4.5.5) in R. Venn diagrams were generated using the web tool Venn diagram (<http://bioinformatics.psb.ugent.be/webtools/Venn/>) and redraw with Inkscape. Expression plots were generated using GraphPad Prism (V.9.1).

GO term enrichment analysis

After obtaining the differential gene expression results, Gene Ontology (GO) analysis was performed using the R TopGO package (v2.50.0). This analysis aimed to identify significantly enriched biological processes, cellular components, and molecular functions among the differentially expressed genes. The file "nveGenes.vienna130208.GO_annotation_141017.txt" was utilized for GO analysis, and it was obtained from the following source: https://figshare.com/articles/dataset/Nematostella_vectensis_transcriptome_and_gene_models_v2_0/807696. This file contains the set of GO-transcript annotations that served as input for TopGO. The algorithm assigns a significance score to each GO term based on the enrichment p -value and the specificity of the term. In this study, the GO analysis was performed separately for the up-regulated and down-regulated genes in each condition (LD and DD) to identify the specific biological processes and molecular functions that are affected by the *NvClk*¹ mutation.

E-box motif enrichment analysis

Sequences for promoter regions (1000kb upstream ATG) of differentially expressed genes were extracted. We manually identified in the list of motif enrichment all the E-box motifs and Circadian E-box motifs. Boxplot were generated using GraphPad Prism version 9.5.1.

Differential Expression Analysis

Differential expression analysis was performed using R (v4.2.2) Bioconductor package, DESeq2 (v1.38.3)⁵¹. Raw read counts were obtained using I used HTSeq-Count (v0.12.4)⁵² and then imported into DESeq2 for normalization and statistical analysis. Differentially expressed genes were identified using the Wald test with an adjusted p-value cutoff of 0.05. The analysis was performed on all the time-points pooled of each genotype per light condition. The output of the analysis includes a list of genes with their log2 fold change, p-value, and adjusted p-value. Volcano-plots were generated using GraphPad Prism version 9.5.1.

HCR v.3 *in situ* hybridization

A custom *NvClk* (NVE2080, amplifier: B3 and B5) and *NvMyhc-st* probe set (NVE14552, amplifier: B5) were generated. We used *zfHcrt* probe set (ZDB-GENE-040324-1, amplifier: B1 and B3) as a negative control. For HCR on *Nematostella* juvenile, several alterations were made to a previously described protocol⁴⁰. Briefly, polyps were plucked and fixed in 4% PFA overnight at 4°C. Polyps were washed 3× in 1× PBS and then dehydrated and permeabilized with 2×5 min washes in 100% methanol. The samples were stored at −20°C overnight. To rehydrate the samples, a series of graded MeOH/PBST washes were used for 5 min each: 75% MeOH:25% 1× PBST, 50% MeOH:50% 1× PBST, 25% MeOH:75% 1× PBST, and finally 2× washes in 100% 1× PBST. To further permeabilize the polyps, samples were incubated in 10 µg/ml Proteinase K diluted in 1× PBST for 10 min. Samples were quickly washed 3× in 1× PBST, and then post-fixed with 4% PFA for 10 min. After post-fixation, samples underwent 3×5 min washes with 1× 2xSSC+0.1% Triton. From now, the following solutions (Pre-hybridization, hybridization and probe wash buffers) were lab-made from the cnidarian-adapted hybridization buffer⁵³. Samples were then pre-hybridized with pre-hybridization buffer at 37°C for 30 min. After pre-hybridization, samples were incubated with 2 pmol of the probe set diluted in hybridization buffer for 16 hr at 37°C. To remove the probe mixture solution, samples were washed 2× for 30 min each with probe wash buffer at 37°C. Samples were washed 2× for 5 min with 5× SSC +0.1% Triton and then treated with probe amplification buffer for 30 min at room temperature. Samples were washed into hairpin amplification buffer containing snap cooled amplifier hairpins and were incubated at room temperature, protected from light, overnight. Samples were then washed with successive 3× 5xSSC +0.1% Triton washes: 2× washes for 15min. Nuclear staining was performed using DAPI 1:1000 in PBST during 1h. Samples were then washed with successive 2× 5xSSC +0.1% Triton washes: 2× washes for 5min. Eventually were slide-mounted in glycerol and stored at 4°C.

Microscopy and Image processing

Samples were imaged using a Zeiss LSM 710 with a x63 oil objective. They were slide-mounted in glycerol. Image manipulation was performed with Fiji⁵⁴. For the double probes *NvClk* imaging (Fig1.B), ROIs were generated from each *NvClk* probes signal and only the ROIs positive for the 2 fluorophores were kept. These ROIs were then used to extract from the original picture the signal considered as true mRNA signal. Figures were then assembled in Inkscape (<http://www.inkscape.org/>)

Data availability

The RNA-seq data reported in this study have been deposited to the Sequence Read Archive (SRA), under accession PRJNA935092. All data supporting the findings of this study are included in the manuscript and its supplementary files or available upon request from the corresponding authors.

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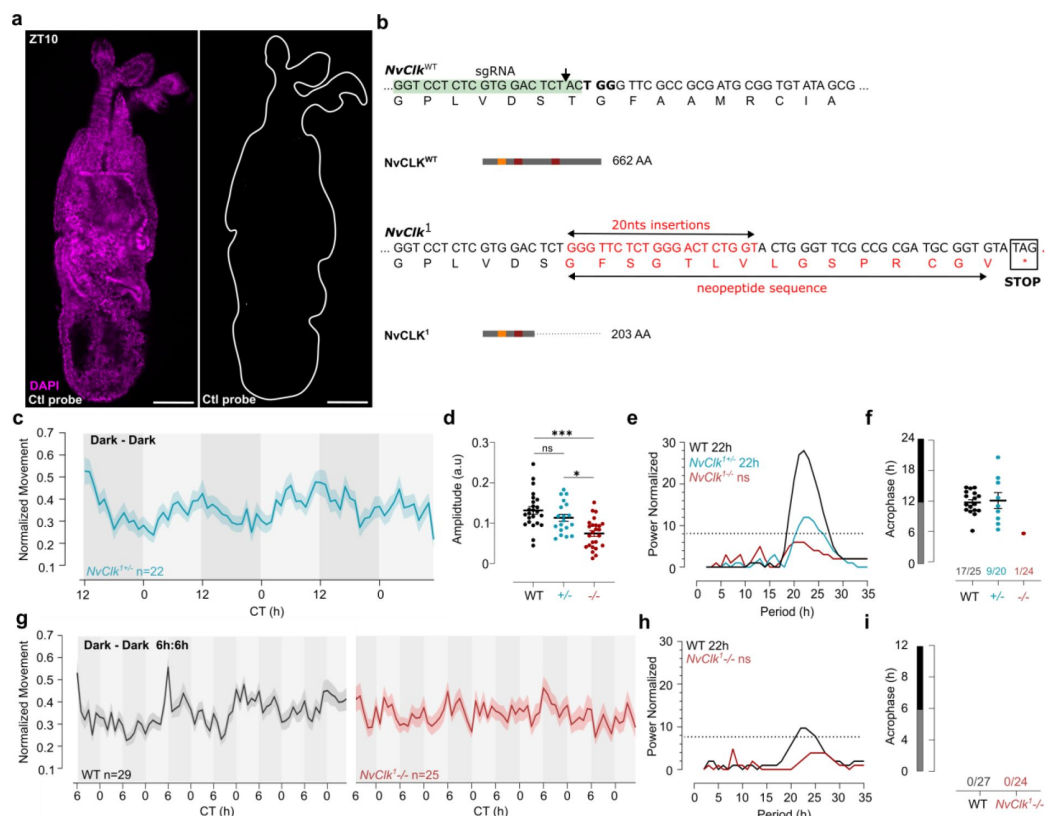
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Author Contributions

M.R., R.A., N.S.B., and O.L. designed the research; M.R., R.A., and N.S.B. carried out the experiments; M.R. and R.A. analyzed the data; T.D. performed the bioinformatics;

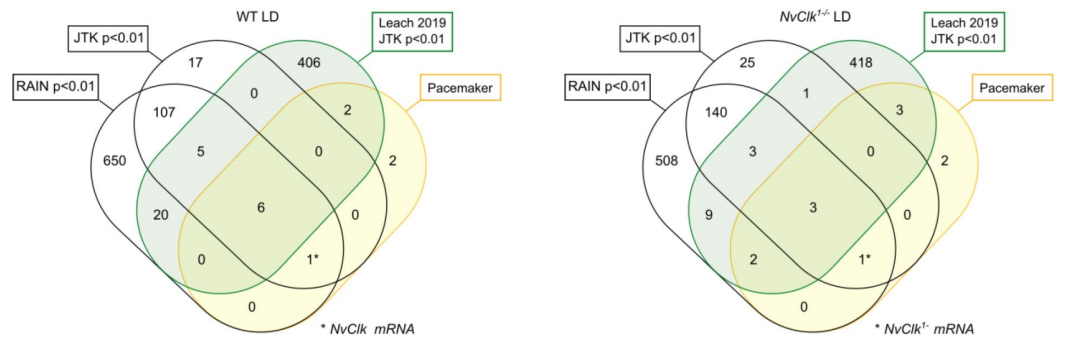
O.L. and L.A. conceived and supervised the project; M.R., R.A., L.A and O.L. wrote the manuscript. All authors read and approved the final manuscript.

No competing Interests



Extended Data Fig. 1

(a) *In situ* hybridization HCRv.3 of control probe (zebrafish transcripts) in the WT juvenile with scale bars representing 0.1mm. **(b)** Nucleotide and translated amino acid sequences of wild-type and *NvCLK*¹ alleles. sgRNA target site is boxed in green, PAM site is in bold, a black arrow indicates predicted site of editing. Predicted translation termination site is boxed in black. Insertion is labeled in red. Predicted immature peptide sequences in red. **(c)** Normalized Movement (a.u.), hourly binned over 72h, under continuous dark (Dark - Dark), Blue line represents the *NvCLK*^{1+/-} heterozygote. **(d)** Individual locomotor amplitude comparison between the three genotype. Kruskal-wallis, multitest comparison *** is $p < 0.001$; * is $p < 0.05$. **(e)** Lomb-Scargle Periodograms for each corresponding genotypes in constant darkness. The significant period value ($p < 0.01$) is indicated for each genotype in the top left corner. **(f)** Phase detection (Cosinor) and genotype comparison of 24h rhythmic individuals. See number nrhythmic/ntotal on the x-axis indicating for each genotype the number of 24h rhythmic animals over the total number analyzed. **(g)** Normalized Movement (a.u.), hourly binned over 72h, under continuous dark (Dark - Dark) after 72h of LD 6h: 6h entrainment, Black line represent the WT and red line represents the *NvCLK*^{1-/-}. **(h)** Lomb-Scargle Periodograms for each corresponding genotypes in constant darkness. The significant period value ($p < 0.01$) is indicated for each genotype in the top left corner. **(i)** Phase detection (Cosinor) and genotype comparison of 24h rhythmic individuals. See number nrhythmic/ntotal on the x-axis indicating for each genotype the number of 24h rhythmic animals over the total number analyzed.



Extended Data Fig. 2

Venn diagram showing overlaps of rhythmic genes in LD (WT on the left and *NvClk*^{1-/-} on the right) condition with rhythmic genes from Leach et al., 2019 and the candidate pacemaker genes (based on protein conservation).

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Reviewer #2 (Public Review):

In this revised manuscript Aguillon and collaborators convincingly demonstrating that CLK is required for free-running behavioral rhythms under constant conditions in the Cnidarian *Nematostella*. The results also convincingly show that CLK impacts rhythmic gene expression in this organism. This original work thus demonstrate that CLK was recruited very early during animal evolution in the circadian clock mechanism to optimize behavior and gene expression with the time-of-day. The manuscript could still benefit from some improvements so that it is more accessible for a wide readership.

<https://doi.org/10.7554/eLife.89499.2.sa0>

Author response:

The following is the authors' response to the original reviews.

***eLife* assessment**

The study provides potentially fundamental insight into the function and evolution of daily rhythms. The authors investigate the function of the putative core circadian clock gene Clock in the cnidarian Nematostella vectensis. While it parts still incomplete, the evidence suggests that, in contrast to mice and fruit flies, Clock in this species is important for daily rhythms under constant conditions, but not under a rhythmic light/dark cycle, suggesting that the major role of the circadian oscillator in this species could be a stabilizing function under non-rhythmic environmental conditions.

Public Reviews:

Reviewer #1 (Public Review):

In this nice study, the authors set out to investigate the role of the canonical circadian gene Clock in the rhythmic biology of the basal metazoan Nematostella vectensis, a sea anemone, which might illuminate the evolution of the Clock gene functionality. To achieve their aims the team generated a Clock knockout mutant line (Clock^{-/-}) by CRISPR/Cas9 gene deletion and subsequent crossing. They then compared wild-type (WT)

with *Clock*^{-/-} animals for locomotor activity and transcriptomic changes over time in constant darkness (DD) and under light/dark cycles to establish these phenotypes under circadian control and those driven by light cycles. In addition, they used Hybridization Chain Reaction-In situ Hybridization (HCR-ISH) to demonstrate the spatial expression of *Clock* and a putative circadian clock-controlled gene *Myh7* in whole-mounted juvenile anemones.

The authors demonstrate that under LD both WT and *Clock*^{-/-} animals were behaviourally rhythmic but under DD the mutants lost this rhythmicity, indicating that *Clock* is necessary for endogenous rhythms in activity. With altered LD regimes (LD6:6) they show also that *Clock* is light-dependent. RNAseq comparisons of rhythmic gene expression in WT and *Clock*^{-/-} animals suggest that clock KO has a profound effect on the rhythmic genome, with very little overlap in rhythmic transcripts between the two phenotypes; of the rhythmic genes in both LD and DD in WT animals (220- termed clock-controlled genes, CCGS) 85% were not rhythmic in *Clock*^{-/-} animals in either light condition. In silico gene ontology (GO) analysis of CCGS reflected process associated with circadian control. Correspondingly, those genes rhythmic in KO animals under DD (here termed neoCCGs) were not rhythmic in WT, lacked upstream E-box motifs associated with circadian regulation, and did not display any GO enrichment terms. 'Core' circadian genes (as identified in previous literature) in WT and *Clock*^{-/-} animals were only rhythmic under entrainment (LD) conditions whilst *Clock*^{-/-} displayed altered expression profiles under LD compared to WT. Comparing CCGs with previous studies of cycling genes in *Nematostella*, the authors selected a gene from 16 rhythmic transcripts. One of these, *Myh7* was detectable by both RNAseq and HCR-ISH and considered a marker of the circadian clock by the authors.

The authors claim that the study reveals insights into the evolutionary origin of circadian timing; *Clock* is conserved across distant groups of organisms, having a function as a positive regulator of the transcriptional translational feedback loop at the heart of daily timing, but is not a central element of the core feedback loop circadian system in this basal species. Their behavioural and transcriptomic data largely support the claims that *Clock* is necessary for endogenous daily activity but that the putative molecular circadian system is not self-sustained under constant darkness (this was known already for WT animals)- rather it is responsive to light cycles with altered dynamics in *Clock*^{-/-} specimens in some core genes under LD. In the main, I think the authors achieved their aims and the manuscript is a solid piece of important work. The *Clock*^{-/-} animal is a useful resource for examining time-keeping in a basal metazoan.

The work described builds on other transcriptomic-based works on cnidaria, including *Nematostella*, and does probe into the molecular underpinnings with a loss-of-function in a gene known to be core in other circadian systems. The field of chronobiology will benefit from the evolutionary aspect of this work and the fact that it highlights the necessity to study a range of non-model species to get a fuller picture of timing systems to better appreciate the development and diversity of clocks.

Strengths:

The generation of a line of *Clock* mutant *Nematostella* is a very useful tool for the chronobiological community and coupled with a growing suite of tools in this species will be an asset. The experiments seem mostly well conceived and executed (NB see 'weaknesses'). The problem tackled is an interesting one and should be an important contribution to the field.

Weaknesses:

I think the claims about shedding light on the evolutionary origin of circadian time maintenance are a little bold. I agree that the data do point to an alternative role for Clock in this animal in light responsiveness, but this doesn't illuminate the evolution of time-keeping more broadly in my view. In addition, these are transcriptomic data and so should be caveated- they only demonstrate the expression of genes and not physiology beyond that. The time-course analysis is weakened by its low resolution, particularly for the RAIN algorithm when 4-hour intervals constrain the analysis. I accept that only 24h rhythms were selected in the analysis from this but, it might be that detail was lost - I think a preferred option would be 2 or 3-hour resolution or 2 full 24h cycles of analysis.

The authors discount the possibility of the observed 12h rhythmicity in Clock^{-/-} animals by exposing them to LD6:6 cycles before free-running them in DD. I suggest that LD cycles are not a particularly robust way to entrain tidal animals as far as we know. Recent papers show inundation/mechanical agitation are more reliable cues (Kwiatkowski ER, et al. Curr Biol. 2023, 2;33(10):1867-1882.e5. doi: 10.1016/j.cub.2023.03.015; Zhang L., et al Curr Biol. 2013, 23;19, 1863-1873 doi.org/10.1016/j.cub.2013.08.038.) and might be more effective in revealing endogenous 12h rhythms in the absence of 24h cues.

Response: We removed the suggestion that we used 6:6h LD to perform tidal entrainment. We generated this ultradian light condition to address the 24h rhythmicity observed in the NvClk1^{-/-} in 12:12h LD.

Reviewer #2 (Public Review):

This manuscript addresses an important question: what is the role of the gene Clock in the control of circadian rhythms in a very primitive group of animals: Cnidaria. Clock has been found to be essential for circadian rhythms in several animals, but its function outside of Bilaterian animals is unknown. The authors successfully generated a severe loss-of-function mutant in Nematostella. This is an important achievement that should help in understanding the early evolution of circadian clocks. Unfortunately, this study currently suffers from several important weaknesses. In particular, the authors do not present their work in a clear fashion, neither for a general audience nor for more expert readers, and there is a lack of attention to detail. There are also important methodological issues that weaken the study, and I have questions about the robustness of the data and their analysis. I am hoping that the authors will be able to address my concerns, as this work should prove important for the chronobiology field and beyond. I have highlighted below the most important issues, but the manuscript needs editing throughout to be accessible to a broad audience, and referencing could be improved.

Major issues:

(1) Why do the authors make the claim in the abstract that CLOCK function is conserved with other animals when their data suggest that it is not essential for circadian rhythms? dCLK is strictly required in Drosophila for circadian rhythms. In mammals, there are two paralogs, CLOCK and NPAS2, but without them, there are no circadian rhythms either. Note also that the recent claim of BMAL1-independent rhythms in mammals by Ray et al., quoted in the discussion to support the idea that rhythms can be observed in the absence of the positive elements of the circadian core clock, had to be corrected substantially, and its main conclusions have been disputed by both Abruzzi et al. and Ness-Cohn et al. This should be mentioned.

Response: According to our Behavioral and Transcriptomic data, CLOCK function is conserved in constant light condition. In LD context, the rhythmicity is maintained probably by the light-response pathway in Nematostella. We modified our rhythmic transcriptomic

analysis and considered the context of the contested results by Ray et al., and discussed it in the revised manuscript.

(2) The discussion of CIPC on line 222 is hard to follow as well. How does mRNA rhythm inform the function of CIPC, and why would it function as a "dampening factor"? Given that it is "the only core clock member included in the Clock-dependent CCGs," (220) more discussion seems warranted. Discussing work done on this protein in mammals and flies might provide more insight.

Response: The initial sentence was unclear. Furthermore, since we restricted our rhythmic analysis to genes only found rhythmic with a $p < 0.01$ with RAIN combined with JTK, NvCipc was no longer defined as rhythmic in free running.

(3) The behavioral arrhythmicity seen with their Clock mutation is really interesting. However, what is shown is only an averaged behavior trace and a single periodogram for the entire population. This leaves open the possibility that individual animals are poorly synchronized with each other, rather than arrhythmic. I also note that in DD there seem to be some residual rhythms, though they do not reach significance. Thus, it is also possible that at least some individual animals retain weak rhythms. The authors should analyze behavioral rhythms in individual animals to determine whether behavioral rhythmicity is really lost. This is important for the solidity of their main conclusions.

Response: Fig. 1 has been modified. We have separated the data for WT and NvClk1^{-/-} animals to provide clarity on the average behavior pattern for each genotype. While the LSP analysis on the population average informs us about the synchronization of the population, it is true that it does not provide insight into individual rhythmicity. To address this, we analyzed individuals in all conditions using the Discorhythm website (Carlucci et al., 2019).

In the revised figure, we have included a comparison plot of the acrophase of 24-hour rhythmic animals between genotypes using Cosinor analysis, which is most suitable for acrophase detection. This plot indicates the number of animals detected as significantly rhythmic, providing direct visual input to the reader regarding individual rhythmicity. Additionally, we have added Table 1, which contains the Cosinor period analysis (24 and 12 hours) of individuals for all genotypes and conditions, further enhancing the clarity of our findings.

(4) There is no mention in the results section of the behavior of heterozygotes. Based on supplement figure 2A, there is a clear reduction in amplitude in the heterozygous animals. Perhaps this might be because there is only half a dose of Clock, but perhaps this could be because of a dominant-negative activity of the truncated protein. There is no direct functional evidence to support the claim that the mutant allele is nonfunctional, so it is important to discuss carefully studies in other species that would support this claim, and the heterozygous behavior since it raises the possibility that the mutant allele acts as a dominant negative.

Response: Extended Data Fig.1 modified. We show NvClk1^{+/-} normalized locomotion over time in DD of the population, comparison of individual normalized behavior amplitude, LSP of the average population and individual acrophase of only rhythmic 24h individuals. Indeed, we cannot discriminate Dominant-negative from non-functional allele.

(5) I do not understand what the bar graphs in Figure 2E and 3B represent - what does the y-axis label refer to?

Response: Not relevant to the revised manuscript.

(6a) I note that RAIN was used, with a $p < 0.05$ cut-off. I believe RAIN is quite generous in calling genes rhythmic, and the p-value cut-off is also quite high. What happens if the stringency is increased, for example with a $p < 0.01$.

Response: We acknowledge your concern regarding the stringency of our statistical analysis. To address this, we opted to combine both RAIN and JTK methods and applied a more stringent p-value cut-off of $p < 0.01$.

(6b) It would be worth choosing a few genes called rhythmic in different conditions (mutant or wild-type. LD or DD), and using qPCR to validate the RNAseq results. For example, in Figure 3D, Myh7 RNAseq data are shown, and they do not look convincing. I am surprised this would be called a circadian rhythm. In wild-type, the curve seems arrhythmic to me, with three peaks, and a rather large difference between the first and second ZT0 time point. In the Clock mutants, rhythms seem to have a 12hr period, so they should not be called rhythmic according to the material and methods, which says that only ca 24hr period mRNA rhythms were considered rhythmic. Also, the result section does not say anything about Myh7 rhythms. What do they tell us? Why were they presented at all?

Response: Regarding the suggestion for independent verification of our RNAseq results, we agree that such validation would enhance the robustness of our findings. To address this, we chose to overlap our identified rhythmic genes under WT LD conditions with those from another transcriptomic study that shared similarities in experimental design. Notably, the majority of overlapping rhythmic genes between the studies are candidate pacemaker genes. We believe that this replication of biologically significant rhythmic genes strengthens the validity and reliability of our results (see Extended Data Fig. 2).

Furthermore, we have decided to remove the NvMhc-st (mistakenly named Myh7, only rhythmic in WT DD in the new analysis) as it does not contribute substantively to the revised version of the manuscript.

(7) The authors should explain better why only the genes that are both rhythmic in LD and DD are considered to be clock-controlled genes (CCGs). In theory, any gene rhythmic in DD could be a CCG. However, Leach and Reitzel actually found that most genes in DD1 do not cycle the next day (DD2)? This suggests that most "rhythmic" genes might show a transient change in expression due to prolonged obscurity and/or the stress induced by the absence of a light-dark cycle, rather than being clock controlled. Is this why the authors saw genes rhythmic under both LD and DD as actual CCGs? I would suggest verifying that in DD the phase of the oscillation for each CCG is similar to that in LD. If a gene is just responding to obscurity, it might show an elevated expression at the end of the dark period of LD, and then a high level in the first hours of DD. Such an expression pattern would be very unlikely to be controlled by the circadian clock.

Response: As we modified our transcriptomic analysis, we do no longer analyze LD+DD rhythmic genes, but any genes rhythmic (RAIN and JTK $p < 0.01$) in each condition. As such we end up with four list of genes corresponding to each experimental conditions.

(8) Since there are still rhythms in LD in Clock mutants, I wonder whether there is a paralog that could be taking Clock's place, similar to NPAS2 in mammals.

Response: see response to (1) > The only NPAS2 orthologous identified in Nematostella NPAS3 showed marginally significance ($p = 0.013$) with RAIN in LD WT suggesting a regulation similar

to the candidate pacemaker genes. As such we included within our candidate pacemaker genes list.

(9) I do not follow the point the authors try to make in lines 268-272. The absence of anticipatory behavior in Drosophila Clk mutants results from disruption of the circadian molecular clock, due to the loss of Clk's circadian function. Which light-dependent function of Clock are the authors referring to, then? Also, following this, it should be kept in mind that clock mutant mice have a weakened oscillator. The effect on entrainment is secondary to the weakening of the oscillator, rather than a direct effect on the light input pathway (weaker oscillators have increased response to environmental inputs). The authors thus need to more clearly explain why they think there is a conservation of circadian and photic clock function.

Response: Following the changes in our statistical analysis we reframed the discussion and address directly the circadian and the photic clock function (we call it light-response pathway in the manuscript)

Recommendations for the authors:

We suggest the following improvements:

(1) Please undertake a serious effort to make this work more accessible to non-marine chronobiologists. This includes better explanations, and schemes of the animal when images of staining are shown (e.g. Fig. 1b) which include the labeling of relevant morphological structures mentioned in the text (like "tentacle endodermis and mesenteries" (line 132)). Similar issues for mentioned life cycle stages like "late planula stage" (line 133), "bisected physa" (line 149).

Response: Fig. 1b, we outlined the animal shaped and added 2 arrows to locate the tentacle endodermis and mesenteries. We replaced the term late planula stage, by larvae. And we rephrased bisected physa by tissue sampling.

Please attend to details. This includes:

- *Wrong referrals to figures (currently line 151 refers to EDF2- but should be EDF 1 instead, there is a Fig.3f mentioned in the text, but there is no such Fig.).*

Response: Fixed

- *Mentioning of ZTs when the HCR stainings were performed.*

Response: Fixed

- *Fig.1 a shows a rather incomplete and thus potentially confusing phylogenetic tree. Vertebrates have at least two Clk orthologs (NPAS2 and CLK), please include both, use an outgroup, and rout the tree.*

Response: Identifying NPAS2 and CLK orthologous in all species added more confusion into the conclusion. However, we followed the suggestion of adding an outgroup using a CLK orthologous sequence identified in the sponge Amphimedon queenslandica and rout the tree. Thank for the suggestion.

- *What do the y-axis labels in Figure 2E and 3B refer to exactly? Y-axis label annotations in Fig.3a,d are entirely missing- what do the numbers refer to?*

Response: not relevant in the revised manuscript

- *Fig.2D- is the Go term enrichment referring to LD or DD?*

Response: to DD. We made it cleared on the figure 5.

- *Wording: "Clock regulates genetic pathways." What is meant by "genetic pathways"? There are no "non-genetic pathways". Could one simply say: "Clock regulates a variety of transcripts".*

Response: We modified our threshold to use only $p_{adj} < 0.01$, which reduced the GO term numbers. We removed “genetic pathways” and now address the specific pathways: cell-cycle and neuronal.

The use of the term "epistatic" is confusing (line 219), i.e. that light is epistatic to Clock. In genetics, epistasis is defined as the effect of gene interactions on phenotypes. To a geneticist, this implies that there is a second gene impacting on the phenotype of the Clock mutants. Please re-word.

Response: “light is epistatic on Clock” has been re-phrased.

The provided Supplementary tables are not well annotated. Several of them need guess-work about what is shown. For instance, for Supplementary Table 1, the Ns are unclear, which in total can go up to almost 200 per condition-genotype, but only about 30 animals for each were tested. Thus, where do the high totals in the LSP table come from? What do the numbers of each periodicity mean? Initially one might assume it was the number of animals that showed a periodogram peak at a given periodicity, but it seems that cannot be. Maybe it counted any period bin over statistical significance? Please clarify with better descriptions and labels.

Response: Supplementary tables are now clearly annotated on their first Tabs. About Fig.1, we already addressed this point in the public review.

Albeit not essential, it would be more reader-friendly to also add a summary table with average period and SD, power and SD, and percentage rhythmicity to the main figure.

Response: Table 1 is added: it contains individual count of rhythmic animals (24h and 12h) with Cosinor. However, using Discorhythm we had to ask for a specific Period. Thus, we can only provide animal count significant for a given period value. And not an estimation of their own period.

(2) Some of the terminology is quite confusing, in particular the double meaning of the word "clock" (i.e the pacemaker and the transcription factor). This is not a specific problem to this manuscript, but it would be helpful for the readability to try to improve this.

Could the gene/transcript/protein be spelled: clk and Clk?

Alternatively, for clarity- how about talking about "core pacemaker genes," "CLOCK-dependent rhythmic genes" and "CLOCK-independent rhythmic genes"?

Response:

Clock/CLOCK > NvClk / NvCLK and the mutant is NvClk1-/-

Core clock genes > candidate pacemaker genes.

CLOCK-dependent CCG > this notion no longer exists in the revised manuscript.

CLOCK-independent CCG > this notion no longer exists in the revised manuscript.

(3) The dismissal of the 12h rhythmicity in Clock^{-/-} animals is not really convincing and should be reconsidered. LD6:6 cycles (before free-running animals in DD) is likely a not particularly robust way to entrain tidal animals. Recent papers show inundation/mechanical agitation are more reliable cues (Kwiatkowski ER, et al. Curr Biol. 2023, 2;33(10):1867-1882.e5. doi: 10.1016/j.cub.2023.03.015; Zhang L., et al Curr Biol. 2013, 23;19, 1863-1873 doi.org/10.1016/j.cub.2013.08.038.) and might be more effective in revealing endogenous 12h rhythms in the absence of 24h cues.

Response: We removed the proposition of using 6:6hLD as Tidal entrainment. Instead, the LD 6:6 experiment reveals the direct light-dependency of the NvClk1^{-/-} mutant.

(4) There are significant questions raised on the validity of BMAL1-independent rhythms in mammals as suggested by the Ray et al study. See DOI: 10.1126/science.abe9230 and DOI: 10.1126/science.abf0922

These technical comments should also be taken into account and the discussion adjusted accordingly to better reflect the ongoing discussions in the chronobiology field.

Response: We modified our rhythmic analysis. As we cannot use BHQ or adjusted p-value which resulted in very genes, we defined 24h-rhythmic genes if $p < 0.01$ with two different algorithms (RAIN and JTK). We propose this compromise to reduce the risk of false-positive. Furthermore, we discussed our methodology in the light of the significant questions raised by these papers you cited. We thank the reviewer for this important point.

(5) The HCR stainings for clk are not very convincing. Normally, HCR should have more dots. In principle, the logic of HCR is such that it detects individual mRNA molecules in the cell. Thus, having only one strong dot/cell like in Fig.1b doesn't make much sense.

Response: We were the first surprised by this single dot signal. We are experienced users of HCRv.3 across different species. We decided to remove the close-up (for further investigations) but to keep the full animal signal. According to our approach it is a convincing signal. However, the doty nature of the signal itself it is not easy to make it highly visible at full scale animal on the picture. We did our best to show the mRNA signal visible without altering the pattern.

Furthermore, the controls for the HCR in situ hybridization are unclear. In the methods, there are two Clock probes described (B3 & B5) and two control probes (B1 & B3), however, in the negative control image, a combination of one Clock (B1) and one control (B3) probes is used and is unclear what "redundant detection" means in the legend of figure S2.

Response: Considering the nature of the signal (single of few dots), we decided to use two probes with 2 different fluorophores. A noise is by nature random. Our hypothesis was: only overlapping fluorescent dots are true signal of NvClk mRNA.

For Control probes we used two zebrafish probes labelling hypothalamic peptides.

Based on the experience with non-Drosophila, non-mouse animal model systems the reviewers assume that non-sense mediated mRNA decay (NMD) is not strongly initiated

upon Crispr-induced premature STOP-codons. If this assumption is correct it would be worth to mention it. Alternatively, it would be worth testing if Nematostella induces NMD, as this would be a great control for the HCR and the mutation itself. At which ZT was the HCR done?

Response: We performed the HCR at ZT10 when NvClk is described to be at peak. It is now indicated in the Fig. 1b. The RNAseq detected a higher quantity of NvClk1 mRNA in the NvClk1-/- (see Fig. 4a). mRNA quantity regulation involves transcription, stabilization, and degradation. At this stage, we cannot identify which specific step is affected.

For Fig.1c- please provide the binding site and sequence in the figure, simply include EDF 1 in the main figure.

Response: We generated a clear indication in the new Fig.1c and EDF. 1b about the protein domains, the CRISPR binding site and the consequences on the DNA and AA sequences.

(6) Please provide the individual trace data for the behavioral analyses either as supplementary files or as a link to an openly accessible database like DRYAD (see also comment 7 in the public review of reviewer 2). Maybe this is what is shown in Supplementary Table 1, but it is really not clear what is actually shown.

Response: Fig.1 is updated. Table 1 is added. Supplementary Table 1 contains individual normalized locomotor data of each polyps for each genotypes and light conditions. Supplementary Table 2 contains the cosinor individual rhythmic behavior analysis based on the Supplementary Table 1.

(7) It is not really clear if the mutation is a true loss-of-function or could also be dominant negative. While this is raised in the discussion, it should be more carefully considered. The reason why a dominant negative would be unlikely is unclear. More specifically also see comment 8) in the public review of reviewer 2.

Response: Indeed, the results cannot tell us if it is a true loss of function, a dominant negative or non-functional allele. We addressed it in the first part of the discussion.

(8) The pretty small overlap of rhythmic transcripts in LD and DD could reflect the true biology of a more core clock driven-process under constant conditions and a more light-driven process under LD. But still- wouldn't one expect that similar processes should be rhythmic? If not, why not?

It would certainly add strength to the data if for one or two transcripts these results were independently verified by qPCR from an independent sampling. This could even be done for just two time points with the most extreme differences.

Response: We appreciate the reviewer's comments and concerns regarding the overlap of rhythmic transcripts in different conditions. In response to the reviewer's query, we revised our interpretation of the transcriptomic data, acknowledging the limited overlap between light and genotype conditions in our study. This prompted us to reconsider the underlying biological processes driving rhythmic gene expression under constant conditions versus light-dark cycles.

Regarding the suggestion for independent verification of our RNAseq results, we agree that such validation would enhance the robustness of our findings. To address this, we chose to overlap our identified rhythmic genes under WT LD conditions with those from another transcriptomic study that shared similarities in experimental design. Notably, the majority of overlapping rhythmic genes between the studies are candidate pacemaker genes. We believe

that this replication of biologically significant rhythmic genes strengthens the validity and reliability of our results (see Extended Data Fig. 2).

(9) *Expression of myh7 : Checking for co-expression should be pretty straightforward by HCR. This is what this type of staining technique is really good for. Please do clk and myh7 co-staining if you want to claim co-expression. Otherwise don't make such a claim.*

Response: We agree that checking for co-expression should be straightforward by HCR. However, due to time constraints during the revision period, we are unable to conduct the double in-situ experiment. Additionally, upon careful consideration, we recognize that including myhc-st (mistakenly named myh7) staining and co-expression analysis would not significantly contribute to the main conclusions of our study. Therefore, we have decided to remove this analysis from the revised manuscript.

(10) *Missing methodological details:*

- *The false discovery rate for each analysis should be included (see Hughes et al.,: "Guidelines for Genome-Scale Analysis of Biological Rhythms," 2017).*

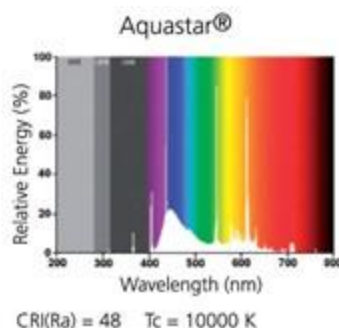
Response: THE FDR is indicated for each gene in supplementary table 3

- *Fig.1f- continuous light- please provide a spectrum (If there is no good spectrophotometer available, please provide at least manufacturer information.*

Response: Unfortunately, we don't have a good spectrophotometer available during the time of the revision. We added to the method the reference of the lamp. We found the light spectrum provided by the supplier. However, we did not add it to the revised manuscript.

Author response image 1.

Spectrum of the Aquastar t8



Also, it would be easier for the reader, if the measurements of light intensity are provided in photons, because this is what the light receptors ultimately measure.

Response: Modified.

- *Fig.2E- please add the consensus sequence used for circadian E-box vs. E-box to the figure.*

Response: In the revised manuscript Fig.4c, we show which E-box motifs we extracted for our promoter analysis. We as well changed our analysis and did no longer use HOMER, but we directly extracted promoter sequences and looked for canonical Ebox CANNTG and Circadian Ebox CACGTG and generate a Circadian Ebox enrichment output per gene promoter.

(11) There has been some discussion about the evolutionary statement as stated by the authors. It appears that depending on the background of the reader, this can be misunderstood. We thus suggest to more clearly point out where the author thinks there is evolutionary conservation (a function for clk in the circadian oscillator under constant light or dark conditions) versus where there is no apparent evolutionary conservation (the situation under light-dark conditions).

Response: In the revised manuscript we proposed a conserved function of NvCLK in constant darkness, and a light-response pathway compensating in LD conditions in the mutant.

Please also consider the major comments 8 and 9 of the common review from reviewer 2.

Reviewer #1 (Recommendations For The Authors):

The hybridization chain-reaction ISH is OK but, I'm not sure I understand the control condition-this should be clarified. I would also welcome the use of Clock^{-/-} animals in HCR as another, more direct level of control. In addition, the authors state that the Myh7 probes hybridise in anatomical regions resembling those for Clock (Fig 3e). It would be better to duplex these two probe sets with different fluors for a better representation of the relative spatial distributions of each transcript.

Response: We agree that checking for co-expression should be straightforward by HCR. However, due to time constraints during the revision period, we are unable to conduct the double in-situ experiment. Additionally, upon careful consideration, we recognize that including myhc-st (mistakenly named myh7) staining and co-expression analysis would not significantly contribute to the main conclusions of our study. Therefore, we have decided to remove this analysis from the revised manuscript.

We clarified in the methods the control probes design.

Minor points:

Figure legends do not all convey sufficient detail. For instance, Figure 1c needs a better explanation. Figure 3e- are these images both WT? Fig 3f doesn't exist and other figure text references do not align with figures and need an overhaul.

Response: All errors have been fixed.

Reviewer #2 (Recommendations For The Authors):

Major issues:

(1) The authors need to introduce their model system better for a broad audience. What are the tissues/cells that express Clock at a higher level? What is their function, does this provide a potential explanation for their specific Clock expression, and how CLOCK might regulate behavior? Terms such as "tentacle endodermis and mesenteries" (line 132), "late planula stage" (line 133), "bisected physa" (line 149) would need some explanation.

Response: We modified term such as planula to larvae, and bisected physa to tissue samples.

1. Some of the terminology used is quite confusing, because of the double-meaning of the word "clock" (i.e the pacemaker and the transcription factor). The authors use terms such as "clock-controlled genes", "core clock genes", "CLOCK-dependent clock-controlled genes", "neo-clock-controlled genes". Is there any way to help the reader? Here are several suggestions: "core pacemaker genes," "CLOCK-dependent rhythmic genes" and "CLOCK-independent rhythmic genes".

Response: all the terminology has been clarified, see previous comments

1. Also in the abstract, there is mention of "hierarchal light- and Clock-signaling" (52-3) - is this related to the statement on line 219 that light is epistatic to Clock? I do not quite understand what epistatic would mean here. Who is upstream of whom? LD modifies rhythmicity in Clock mutant animals, but Clock mutations also impact rhythmicity in LD. Also, as epistasis is defined as the effect of gene interactions on phenotypes - what is the secondary gene impacting the phenotype of the Clock mutants? I am not sure the term epistatic is appropriate in the present context.

Response: Indeed, Epistatic is a genetic term which might be unclear in this context. We removed it.

1. The control for the *in situ* hybridization is unclear. In the methods, there are two Clock probes described (B3 & B5) and two control probes (B1 & B3), however, in the negative control image, a combination of one Clock (B1) and one control (B3) probe is used, I am not sure what "redundant detection" means in the legend of figure S2. Also, the sequences of each Clock probe should be provided. It might be worth testing the Clock mutant the authors generated. Clock mRNA could be reduced due to non-sense, mediated RNA decay, since the mutation causes a premature stop codon. This would be a great additional control for the *in situ* hybridization. Even better would be if, by chance, the probes target the mutated sequence. The signal should then be completely lost.

Response: HCR is a tilling probe. Which means the target transcript is covered by dozens of successive DNA sequence "primer-like" which allow the HCRv.3 technology. We cannot design a mutant probe specific with this technology.

(5) I have concerns with rhythmic-expression calls, particularly as there is so little overlap between LD and DD, and that a completely different set of rhythmic genes is observed in Clock mutant and wild-type animals. I am not an expert in whole-genome expression studies, so I hope one of my colleague reviewers can weigh in.

When describing rhythmicity analysis in the Methods, it states that Benjamini-Hochberg corrections were applied to account for multiple comparisons. However, the false discovery rate for each analysis should be included (see Hughes et al., "Guidelines for Genome-Scale Analysis of Biological Rhythms," 2017).

Response: As explained before we cannot used Benjamini-Hochberg corrections as only few genes (mostly oscillator gene pass the threshold). As such we combined two different algorithms (RAIN and JTK) with a $p < 0.01$ to detect confidently rhythmic genes while reducing the risk of false-positives.

Minor issues:

(1) Environmental inputs are not "circadian", as written in the title.

Response: Title modified

| (2) *In the abstract, the description of the Clock mutant behavioral phenotypes is hard to follow, with no mention of whether or not Clock mutant animals are behaviorally rhythmic or arrhythmic in constant conditions.*

Response: corrected

| (3) *Abstract: A 6/6 h LD cycle is not a compressed tidal cycle as written in the abstract. Light is not an input to tidal rhythms.*

Response: corrected

| (4) *Line 101: timeout is not a core clock gene in animals.*

Response: we removed it from the candidate pacemaker genes.

| (5) *What is the evidence for the role of PAR-Zip proteins in the Nematostella clock? The reference provided does not mention those.*

Response: There is no functional data in *Nematostella* yet to support their role within the pacemaker. However based on their rhythmicity in LD and protein conservation, we included them within the candidate pacemaker genes list. The references have been corrected.

| (6) *Line 125. should refer to Fig 1C when describing the Clock protein.*

Response: corrected

| (7) *Line 143-4. based on the figure, the region targeted by gRNA was not "close to the 5' end" as stated, it is closer to the middle of the gene sequence as shown in Figure 1C. A more accurate description would be a region in between the PAS domains.*

Response: Indeed we modified the figure and the text.

| (8) *Line 150. The mutant allele is described as Clock1 initially, then for the rest of the paper as Clock-. Since it is not clear that the allele is a null (see major comment #8), Clock1 should be used throughout the manuscript.*

Response: the allele is named NvClk1 in the revised manuscript

| (9) *Figure 2A, the second CT/ZTO is misplaced.*

Response: Fig. 2 modified in the revised manuscript

| (10) *Figure legend for 2E and 3B. "The 1000bp upstream ATG" is unclear. I guess it means that 1000bp upstream of the putative initiation codon was used.*

Response: Right, and in the revised version we analyzed 5kb upstream the putative ATG.

| (11) *Line 164. The authors write "We discovered..." , but wasn't it already known that these animals are behaviorally rhythmic?*

Response: Fixed

(12) *It would be worth mentioning in the results section the reduced amplitude of rhythms in LL compared to DD (in WT and seemingly also in Clock mutants).*

Response: Indeed, we observed a significant reduction in the mean amplitude in the NvClk1^{-/-} in DD and LL compared WT and NvClk1^{-/-} in LD, DD and LL. However, as rhythmicity is lost by virtually all mutants in LL and DD we do not think these results add to the current interpretation of the gene function.

(13) *Please correct the figure numbers in the main text, there are several mistakes.*

Response: Done

(14) *Line 196, most genes in the quoted study did not cycle on day 2, so whether they are truly clock controlled is questionable.*

Response: We agree, identifying free-running cycling genes in cnidarian remains a challenge to overcome. One of the limitations of this study was to detect rhythmic genes in LD which conserved rhythmicity in DD. However, considering different transcriptomic studies (cited in the discussion) it seems that in the cnidaria phyla rhythmic genes in LD are not necessarily the one we identified rhythmic in DD.

(15) *Line 204-206 needs to be rephrased. It is confusing.*

Response: rephrased

(16) *Line 216. Rephrase to something like: "A similar finding was made for."*

Response: rephrased

(17) *"Clock regulates genetic pathways" sounds quite odd. Do you mean it regulates preferentially specific genetic (or maybe better, molecular) pathways?*

Response: rephrased

(18) *Figure 4 and legend: Dashed lines indicating threshold are missing. Do the black and red dots represent WT and Clock^{-/-}, as indicated in the legend, or up/down, as indicated in the figures?*

Response: Fig.5 modified accordingly. Colors in the Volcano plot indicate Up- (black) versus Down- (red) regulated. It is now coherent within the figure.

(19) *Legend for Extended figure 1. "Immature peptide sequence" is incorrect.*

Response: rephrased

(20) *Extended data Figure 4. What the asterisks labels is unclear.*

Response: EDF4 was modified and become EDF2 with different content. The * indicates NvClk mRNA

(21) *Line 228. Gene "isoforms". I guess the authors mean "paralogs".*

Response: corrected.

(22) Line 232-3/Figure 3e. Please include a comparable image of the Clk ISH to facilitate the comparison of the spatial expression pattern. In addition, where and what is the "analysis" referred to - "the spatial expression pattern of Myh7 closely resembled that of Clock, as evidenced by our analysis"?

Response: the analysis has been removed from the revised manuscript because we currently cannot perform the double ish.

(23) Line 282-3. As mentioned above, it is difficult to be sure that circadian behavior is lost, if only looking at a population of animals.

Response: Fig.1 corrected

(24) Line 301-5. Rephrase.

Response: Rephrased

(25) Line 325. I am not convinced that the author can say that their mutant is amorphic. See Major comment 8.

Response: corrected.

(26) Line 351 "simplifying interactions with the environment". Please explain what is meant here.

Response: this confusing sentence has been removed from the revised manuscript